Simultaneous AI-Assisted Shared Control for Alternative XR Input: A Climbing Testbed

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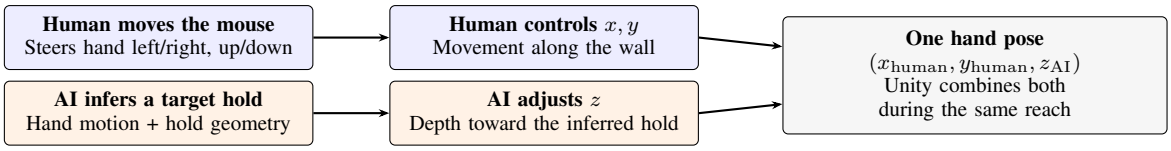


Fig. 1. Two contributions, one reach: the human steers along the wall while AI supplies depth. Grabbing and override remain human-controlled.

***Abstract*—We explore simultaneous human–AI shared control to reduce the controls a person must manage during extended reality (XR) interaction. In a virtual climbing testbed, the person controls lateral movement and grabbing while heuristic assistance adjusts depth. Simulations and exploratory headset testing with one developer reveal tradeoffs among assistance availability, unwanted depth movement, and additional movement. The developer reported lower depth effort with assistance, but consistent performance gains remain unestablished. The work demonstrates how complementary controls can operate during the same action and informs future adaptation of alternative input devices across XR applications. Broader workload and accessibility benefits require further evaluation.**

***Index Terms*—Extended reality, alternative input, shared control, intent inference.**

I. Introduction

A mouse supplies planar movement, but reaching a hold in our extended reality (XR) climbing task also requires depth adjustment. Managing both alongside grabbing adds controls for the person to coordinate. We ask: *How can a person and AI control complementary parts of an XR action simultaneously?* We explore depth assistance while the person steers laterally and grabs, aiming to reduce simultaneous control demands. AI here means heuristic intent inference with deterministic control.

Assigning different axes does not determine which hold assistance should approach or when it should move. Shared-control research connects goal prediction with assistance arbitration [1]–[3]; motion provides pointing cues [4], but misprediction has costs [5]. Assistance must also respond when human movement stops. Incorrect assistance can move toward an unintended hold; conservative assistance can leave depth unfinished.

Using simulations, integration tests, and two headset sessions with one developer, we compare manual, nearest-hold, and intent-based depth control. We examine tradeoffs among assistance availability, unwanted depth movement, and additional human movement. This exploratory study supports the long-term goal of alternative devices performing common XR actions across applications; broader workload and accessibility benefits remain unestablished.

II. Methods

A. Testbed and Input

The person remains seated while controlling a virtual climbing task built in Unity using OpenXR [6] and the XR Interaction Toolkit [7], connected through Meta Quest Link. The environment has no falling or momentum; upward movement is capped at 0.25 m/s, and a visual vignette is applied [8].

Mouse movement positions the active hand along the wall. Pressing the grab button attaches the hand to a reachable hold; dragging downward then raises the virtual body. Releasing the button leaves that hand attached and switches control to the other hand. An unsuccessful grab preserves existing support. The grab action uses the left mouse button, with Space used in earlier runs. E/Q provide manual depth adjustment at 0.15 m/s.

B. Simultaneous Control and Authority

The division of control is fixed: the person controls lateral hand movement, grabbing, and climbing, while assistance adjusts reaching depth. Unity supplies the Python service with hand positions, the active hand, reaching and gripping states, and hold geometry. The service uses this information to infer a target hold and calculate bounded depth adjustments. Images, language, and experimental target cues are not used for inference.

During each Unity update, human-controlled lateral coordinates (x, y) are combined with the accepted assisted depth coordinate z to form one hand pose (Fig. 1). The person can continue moving the mouse while assistance updates depth. Both therefore contribute during the same reach, even though the service and Unity update asynchronously.

The person retains control over grabbing and depth override. Grabbing freezes depth, and pressing E/Q transfers depth control to the person for the remainder of that reach. Unity rejects stale or invalid assistance updates and handles interruptions locally; resuming after an interruption requires deliberate user action.

C. Depth Target and Conditions

Hand contact uses a 0.04 m-radius sphere; holds protrude 0.04, 0.08, or 0.12 m. Assistance targets near-surface contact

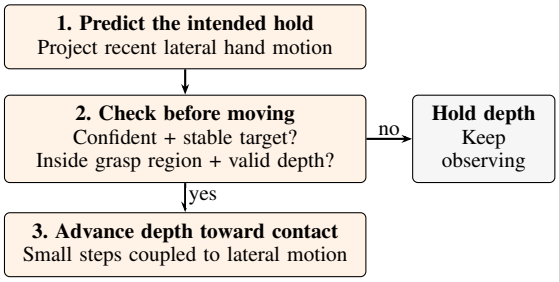


Fig. 2. Intent-assisted depth (nearest has no confidence gate). Uncertainty holds depth. After lateral motion stops, the depth budget ends after 150 ms.

within a hold's lateral grasp region, where valid depth exists. Grabs accept a broader overlap interval; retained depth can permit the next grab without adjustment.

Conditions are *manual* (M; E/Q), *assist-nearest* (N; the nearest unoccupied hold whose lateral grasp region contains the hand), and *assist-intent* (I; motion-inferred target). Both assisted conditions share a depth controller; intent adds the selection rules below.

D. Intent Inference and Target Selection

At 10 Hz, intent estimates lateral velocity $\vec{v}$ over 250 ms, caps speed at 0.3 m/s, and projects lateral position $\vec{q}$ to $\hat{\vec{q}} = \vec{q} + (0.5\,\mathrm{s})\vec{v}$. For unoccupied hold i with lateral centre $\vec{c}_i$ and grasp radius r_i, predicted distance outside its grasp region is $d_i = \max(0, \|\hat{\vec{q}} - \vec{c}_i\| - r_i)$. Weights $w_i = \exp(-d_i^2/2\sigma^2)$, with $\sigma = 0.08$ m, and a "none" weight of 0.1 are normalized to scores p_i.

Selection requires score $p_i \geq 0.80$, a 0.25 lead over the runner-up, and valid grasp geometry and depth, maintained for 200 ms. A selected target's score below 0.55 holds depth; persistence for 200 ms withdraws the target. Invalid geometry withdraws it immediately (Fig. 2). These rules follow prediction and arbitration [2], [9]; scores are uncalibrated, and thresholds and hysteresis are design choices.

E. Motion-Coupled Depth Control

Motion coupling limits both assisted conditions using lateral hand speed. For eligible target depth z^* and current depth z_t, the maximum step b_t and commanded change are

$$\begin{aligned} b_t &= \min(0.015\,\mathrm{m},\ 0.6\,\mathrm{m/s} \cdot \Delta t,\ k\tilde{v}_t \Delta t), \\ \Delta z_t &= \mathrm{sgn}(z^* - z_t)\min(|z^* - z_t|, b_t). \end{aligned} \tag{1}$$

Here Δt is the observation interval and $k = 1.5$. Effective speed $\tilde{v}_t$ uses lateral speed at or above 1 cm/s, retaining the last such speed for a 0.15 s grace period, then zero. Assistance thus continues briefly after stopping, then holds depth. Steps never exceed 1.5 cm, within the client's 2 cm limit. Earlier coupled runs used $k = 0.75$ and a 0.15 m/s cap.

F. Evaluation Design and Measures

Table I summarizes simulations, integration trials, and headset sessions examining controller tradeoffs, signal delivery, and depth effort, respectively. The intent sweep served both tuning and evaluation; results are descriptive.

TABLE I
EVIDENCE SCOPE. CAPS ARE ASSISTED SPEEDS IN M/S; A CELL IS ONE CONDITION AND SETTING.

Evidence	Configuration	Sample / purpose
HTTP integration	Uncoupled; cap 0.15	9 trials; reliability
Intent sweep	Uncoupled; cap 0.15; vary h, σ	400 reaches/setting
Coupling test	Off/on; $k = 0.75$, cap 0.15	400 reaches per cell
Speed comparison	$k = 0.75$, cap 0.15 vs. $k = 1.5$, cap 0.6	120 reaches per cell
Headset sessions	$k = 1.5$, cap 0.6	1 tester; 15 + 18 completed trials

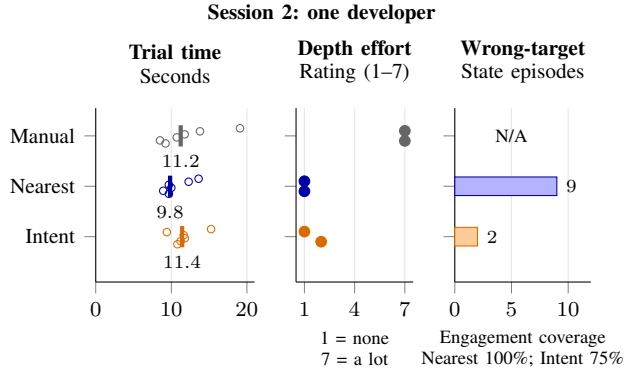


Fig. 3. Complete session 2; six trials/condition. Time: trials (circles), medians (bars and labels). Effort: two block ratings. Marks are offset vertically. Episodes are state-based totals; manual has no AI. Engagement is median trial coverage. Lower counts accompany less engagement.

One developer tested two headset sessions. Each planned one practice trial/condition and six three-trial blocks: M-N-I-I-N-M, then N-I-M-M-I-N. Trials required four cued holds; only completed measured trials enter separate session summaries. Block depth effort used "Getting the depth right took effort" (1: none; 7: a lot). Discomfort ≥ 4 or no rating within 60 s stopped the session.

Trial time spans ready to finish. Assistance-state episodes span entry to a state/target change and are scored against each overlapping reach/cue interval; wrong-target episodes select a non-cued hold. Headset *engagement coverage* is the fraction of intervals containing an episode, summarized by the trial median. States need not produce movement. Wrong-hold travel measures applied depth changes toward non-cued holds. Simulation coverage averages within-reach time assisted toward the cue. Causal checks flag unauthorized movement or direction/cap violations.

Simulated *extra motion* is lateral movement added after lateral arrival and a depth stall of about 0.3 s. The headset metric requires a completed cued reach with a logged coupling hold of at least 0.3 s followed by resumed assistance.

III. RESULTS (PARTIAL)

A. Concurrent Control in Integration Runs

The service and player completed nine scripted HTTP trials without refused or lost signals, causal-motion violations, or data loss. At 72 frames/s, observation-to-application latency had medians of 16–40 ms and a 95th percentile of 42 ms.

B. Intent Inference and Assistance Availability

Compared with nearest, the chosen setting ($h = 0.5$, $\sigma = 0.08$) about halved false-assist time toward distractor P (0.511 vs. 1.006 s per reach crossing P). The cost was lower coverage (0.327 vs. 0.387) and a longer median reach-start-to-first-correct-assistance time (2.01 vs. 1.67 s; among reaches receiving correct assistance). These parameters were selected on the same sweep; the comparison is exploratory.

C. Coordinating Human Motion and Assisted Depth

During 1 s hovers over P (75 per condition), depth travel *after* the 0.15 s grace was 110.5 mm (nearest) and 123.7 mm (intent) without coupling, and zero with it. Coupled nearest still moved 21.7 mm during the grace. In 400 reaches/condition, coupling required extra scripted movement on 32% (nearest) and 77% (intent), versus 0% and 9% without coupling.

In the separate speed comparison, raising k and the cap reduced extra-motion rates from 30% to 17% (nearest) and 72% to 53% (intent). Mean wrong-hold travel rose from 49.7 to 83.1 mm per reach (nearest) and 18.2 to 33.1 mm (intent): faster recovery increased misdirected travel.

D. Headset Testing (Developer)

After a pilot stopped at discomfort 4, interrupted session 1 completed 15 measured trials (M: 3, N: 6, I: 6), excluding three incomplete attempts. Session 2 completed 18/18 (six/condition) plus three practice trials, without data loss, causal-motion violations, or no-signal pauses. All measured completions used the final configuration.

Median trial times (M/N/I) were 36.5/11.7/13.4 s in session 1 and 11.2/9.8/11.4 s in session 2 (Fig. 3). Wrong-target assistance episodes (N/I, session totals) were 21/7 and 9/2, respectively. In session 2, median wrong-hold depth travel was 39.6/0 mm per trial, but maximum single-reach travel was 185.8/93.3 mm. Extra motion was required on 0/24 nearest and 1/24 intent cued reaches.

The developer reported lower effort with assistance (1–2 versus 7), discomfort 1, no clear nearest/intent difference, and no stereo artifacts after fixes.

IV. Discussion (Partial)

Concurrent lateral and depth control lets both contribute during one reach. Assistance handles depth but must select a hold and time its movement to limit motion toward unintended targets.

Intent's fewer wrong-target episodes accompanied lower engagement, not necessarily better inference. Coupling stopped depth after the grace period but could leave it unfinished. Higher gain and caps reduced additional scripted movement while increasing wrong-hold travel.

The developer reported lower depth effort, but session 2 showed little timing advantage. Retained depth and overlap-based grabs allowed 8/24 manual reaches without depth input. Practice, order, and software changes limit comparison; perceptible intent benefits remain unestablished.

Intent required extra motion on 53% of simulated reaches but 1/24 in session 2. Different movement policies and scoring prevent direct comparison or prediction of a human rate.

Evidence comes from one developer and simulations sharing tuning and evaluation data. Gain and cap changed together; manual and assisted caps differed. Wrong-hold travel measures disagreement, not harm; control authority does not establish perceived agency. Accessibility benefits and transfer to other XR actions remain untested.

V. Conclusion

This climbing testbed demonstrates simultaneous shared control: the person controls lateral movement and grabbing while heuristic assistance adjusts depth. The experiments reveal a tradeoff between providing assistance, limiting unwanted depth movement, and requiring additional movement. One developer reported lower depth effort, but consistent performance gains and benefits of intent over nearest assistance remain unestablished. Enabling alternative devices to perform common XR actions across applications remains the long-term goal.

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CRediT Author Statement

Tongda Yin: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Prerna Khanna, Momona Yamagami, and Robert LiKamWa:** Supervision and mentorship.

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